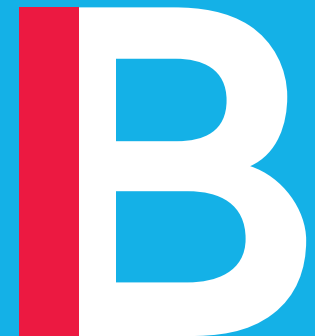


# Quantitative Methods

Difference-in-Difference

by Professor Shuang Zhang



[illegible]

- n the final exam will be done
- s will be posted after the

# Recap: How is RD different from IV?

- RD uses a regression like

$$Y_a = \alpha + \beta D_a + \gamma a + \varepsilon_a$$

- **Assumption**:  $D_a$  is solely determined by  $a$
- By controlling for  $a$ , NO omitted variable correlated with  $D_a$  in the error term

- IV uses two regressions like

$$Y_i = \alpha + \rho Z_i + \varepsilon_i \text{ (Reduced form)}$$

$$D_i = \alpha + \phi Z_i + \mu_i \text{ (First stage)}$$

- $D_i$  is determined by  $Z_i$ , but it doesn't have to be solely determined by  $Z_i$
- **Assumption**:  $Z_i$  is uncorrelated with the error term
  - Stronger assumption than RD

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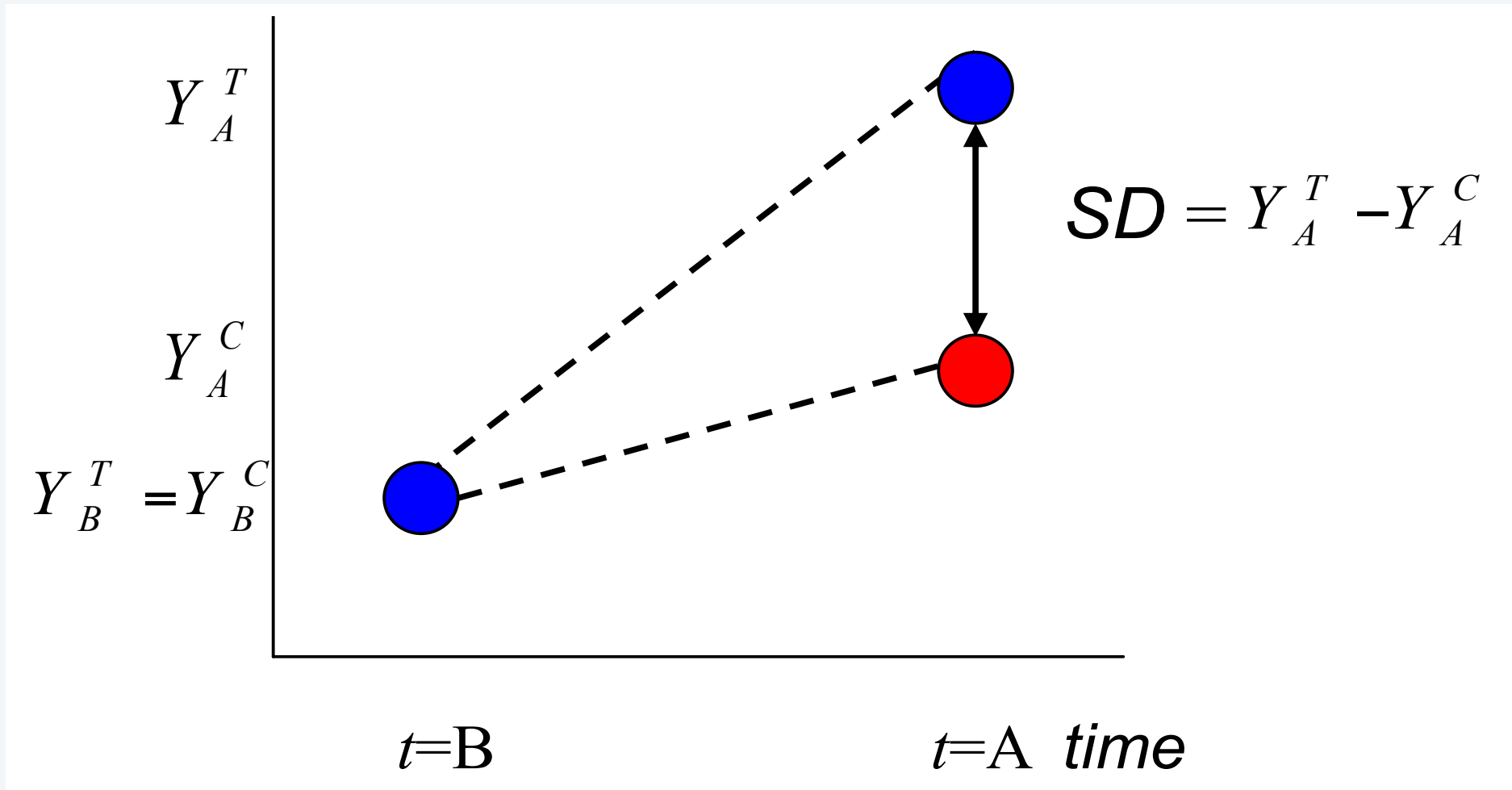
- $D_i$  is determined by  $Z_i$ , but it doesn't have to be solely determined by  $Z_i$
- **Assumption**:  $Z_i$  is uncorrelated with the error term
  - Stronger assumption than RD
- What if neither assumption holds, and your control group and treatment group are indeed different? – ***not the end of the day***

Single difference: difference in group means after intervention

|                               | <b>Treatment Group<br/>With Program</b> | <b>Control Group<br/>Without Program</b> | <b>Difference Across<br/>Groups</b>         |
|-------------------------------|-----------------------------------------|------------------------------------------|---------------------------------------------|
| <b>After<br/>Intervention</b> | Average of outcome<br>$Y_A^T$           | Average of outcome<br>$Y_A^C$            | <b>Single-Difference</b><br>$Y_A^T - Y_A^C$ |

- T: treatment group; C: control group
- A: after intervention; B: before intervention

Single difference: difference in group means after intervention  
(For example, group in means from RCT)



**Strong assumption:**  
T and C are exactly the same  
before the intervention.

However, they are often different.

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## Difference-in-Difference (DID) – basic idea

|                                   | <b>Treatment Group<br/>With Program</b> | <b>Control Group<br/>Without Program</b> | <b>Difference Across<br/>Groups</b>                             |
|-----------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------------------------------|
| <b>After<br/>Intervention</b>     | Average of<br>outcome<br>$Y_A^T$        | Average of outcome<br>$Y_A^C$            | <b>Single-Difference</b><br>$Y_A^T - Y_A^C$                     |
| <b>Before<br/>Intervention</b>    | Average of<br>outcome<br>$Y_B^T$        | Average of outcome<br>$Y_B^C$            | $Y_B^T - Y_B^C$                                                 |
| <b>Difference<br/>across time</b> | $Y_A^T - Y_B^T$                         | $Y_A^C - Y_B^C$                          | <b>Double-Difference</b><br>$(Y_A^T - Y_A^C) - (Y_B^T - Y_B^C)$ |



# The first example in London



**1849:** London's worst cholera epidemic claims 14,137 lives

- Two companies supplied water to much of London: the Lambeth Waterworks Co. and the Southwark and Vauxhall Water Co.
  - ▶ Both got their water from the Thames

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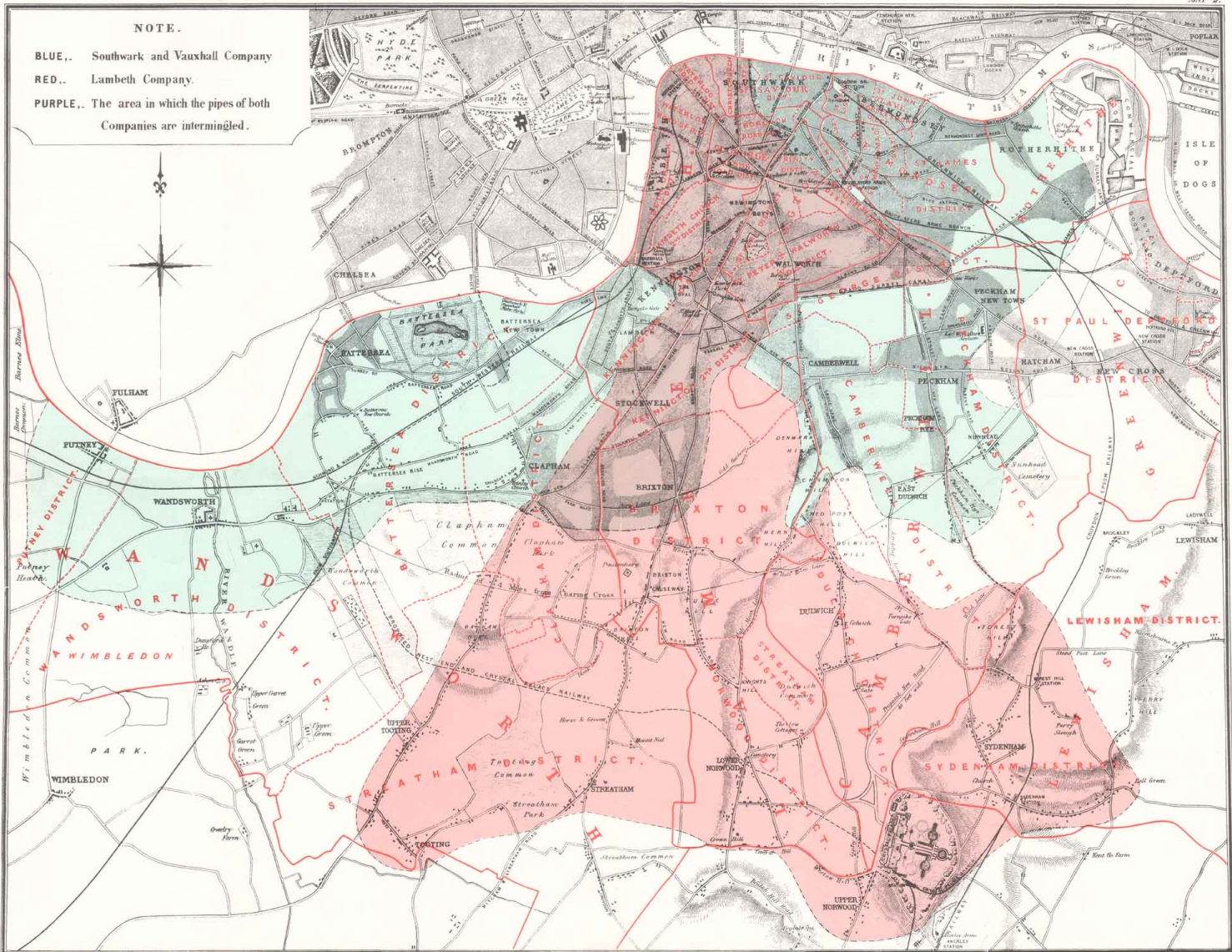
**1852:** Lambeth Waterworks moved their intake upriver

- Everyone knew that the Thames was dirty below central London

**1853:** London has another cholera outbreak

- Are Lambeth Waterworks customers less likely to get sick?

# The first example in London





| Sub-Districts.           | Deaths<br>from<br>Cholera<br>in 1849. | Deaths<br>from<br>Cholera<br>in 1854. | Water Supply.                                            |
|--------------------------|---------------------------------------|---------------------------------------|----------------------------------------------------------|
| St. Saviour, Southwark . | 283                                   | 371                                   | Southwark & Vaux-<br>hall Company only.                  |
| St. Olave . . . .        | 157                                   | 161                                   |                                                          |
| St. John, Horsleydown .  | 192                                   | 148                                   |                                                          |
| St. James, Bermondsey .  | 249                                   | 362                                   |                                                          |
| St. Mary Magdalen . .    | 259                                   | 244                                   |                                                          |
| Leather Market . . .     | 226                                   | 237                                   |                                                          |
| Rotherhithe* . . . .     | 352                                   | 282                                   |                                                          |
| Wandsworth . . . .       | 97                                    | 59                                    |                                                          |
| Battersea . . . . .      | 111                                   | 171                                   |                                                          |
| Putney . . . . .         | 8                                     | 9                                     |                                                          |
| Camberwell . . . . .     | 235                                   | 240                                   |                                                          |
| Peckham . . . . .        | 92                                    | 174                                   |                                                          |
| Christchurch, Southwark  | 256                                   | 113                                   | Lambeth Company,<br>and Southwark and<br>Vauxhall Compy. |
| Kent Road . . . . .      | 267                                   | 174                                   |                                                          |
| Borough Road . . . .     | 312                                   | 270                                   |                                                          |
| London Road . . . . .    | 257                                   | 93                                    |                                                          |
| Trinity, Newington . .   | 318                                   | 210                                   |                                                          |
| St. Peter, Walworth . .  | 446                                   | 388                                   |                                                          |
| St. Mary, Newington . .  | 143                                   | 92                                    |                                                          |
| Waterloo Road (1st) . .  | 193                                   | 58                                    |                                                          |
| Waterloo Road (2nd) . .  | 243                                   | 117                                   |                                                          |
| Lambeth Church (1st) . . | 215                                   | 49                                    |                                                          |
| Lambeth Church (2nd) . . | 544                                   | 193                                   |                                                          |
| Kennington (1st) . . .   | 187                                   | 303                                   |                                                          |
| Kennington (2nd) . . .   | 153                                   | 142                                   |                                                          |
| Brixton . . . . .        | 81                                    | 48                                    |                                                          |
| Clapham . . . . .        | 114                                   | 165                                   |                                                          |
| St. George, Camberwell   | 176                                   | 132                                   |                                                          |
| Norwood . . . . .        | 2                                     | 10                                    | Lambeth Company<br>only.                                 |
| Streatham . . . . .      | 154                                   | 15                                    |                                                          |
| Dulwich . . . . .        | 1                                     | —                                     |                                                          |
| Sydenham . . . . .       | 5                                     | 12                                    |                                                          |
| First 12 sub-districts . | 2261                                  | 2458                                  | Southwk. & Vauxhall.                                     |
| Next 16 sub-districts .  | 3905                                  | 2547                                  | Both Companies.                                          |
| Last 4 sub-districts .   | 162                                   | 37                                    | Lambeth Company.                                         |

## Impact of clean water on reducing deaths

### 2-by-2 DID table

|                      | Number of deaths |       |            |
|----------------------|------------------|-------|------------|
|                      | 1849             | 1854  | Difference |
| Southwark & Vauxhall | 2261             | 2458  | 197        |
| Lambeth              | 162              | 37    | -125       |
| Difference           | -2099            | -2421 | -322       |

The memorial pump in Soho: the handle of which John Snow had removed.

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# Example: Monetary policy during the great depression in 1933

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- Problem during economic downturn: bank failures
- Question: Which monetary policy from the central bank is going to help banks survive?
  - Easy money: lend to banks to help them stay in business
  - Tight money: restrict lending to banks
- Different policies in two regional Federal Reserve Banks from Dec 1930
  - Six district: easy money
  - Eighth district: tight money

# Difference-in-Difference design

- Two groups
  - Treatment group: Six district
  - Control group: Eighth district
- Two time periods
  - Before policy: 1930
  - After policy: 1931
- Even better, half of **Mississippi** belongs to the sixth, and the other half belongs to the eights
  - Same business environment
  - The only difference is different monetary policy

# Effect of easy money on bank survival?

|            | Number of banks |      |
|------------|-----------------|------|
|            | 1930            | 1931 |
| District 6 | 135             | 121  |
| District 8 | 165             | 132  |

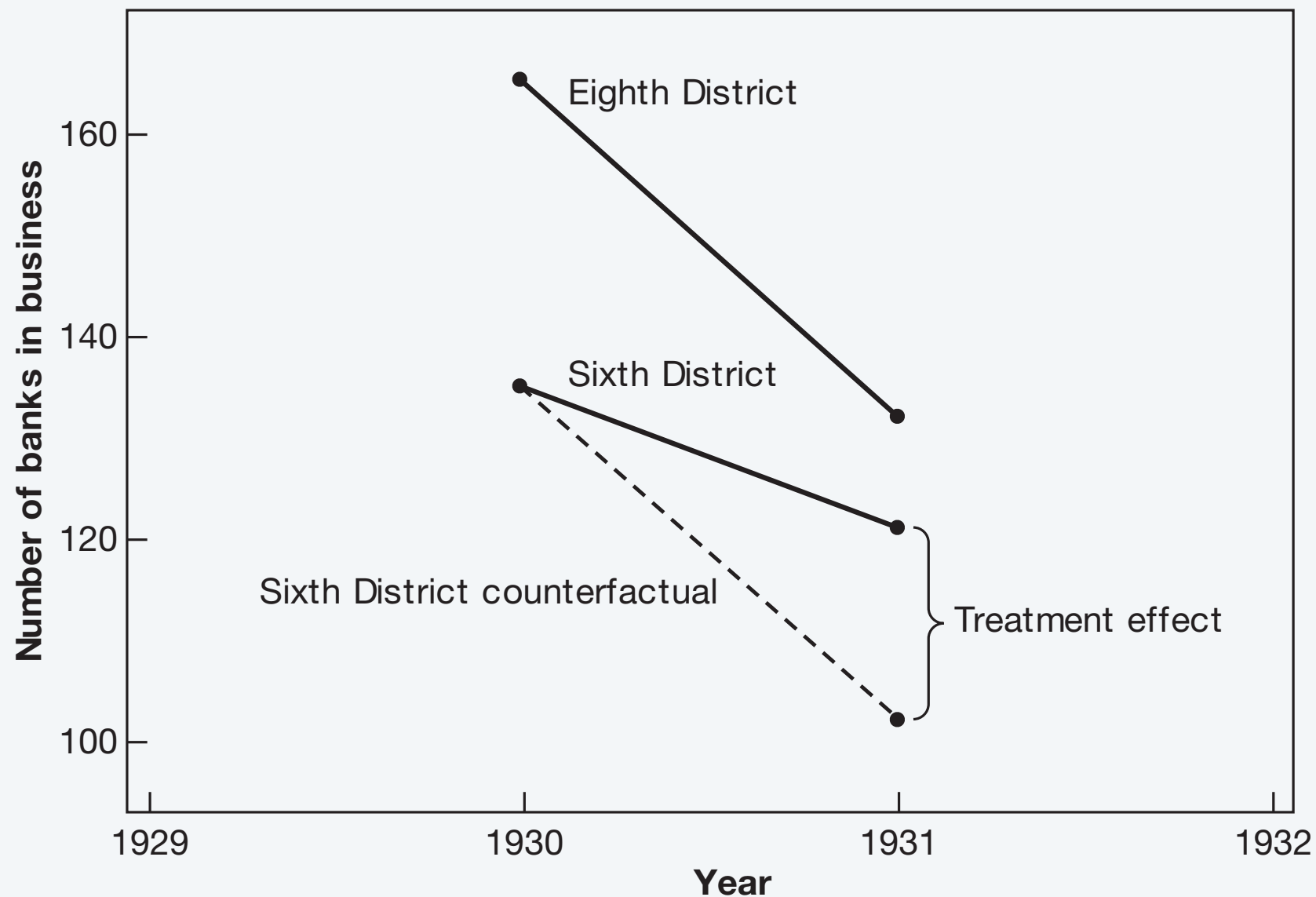
# Effect of easy money on bank survival?

## 2-by-2 DID table

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|            | Number of banks |      | Difference |
|------------|-----------------|------|------------|
|            | 1930            | 1931 |            |
| District 6 | 135             | 121  | -14        |
| District 8 | 165             | 132  | -33        |
| Difference | -30             | -11  | 19         |

FIGURE 5.1  
Bank failures in the Sixth and Eighth Federal Reserve Districts



# Difference-in-Difference: assumption

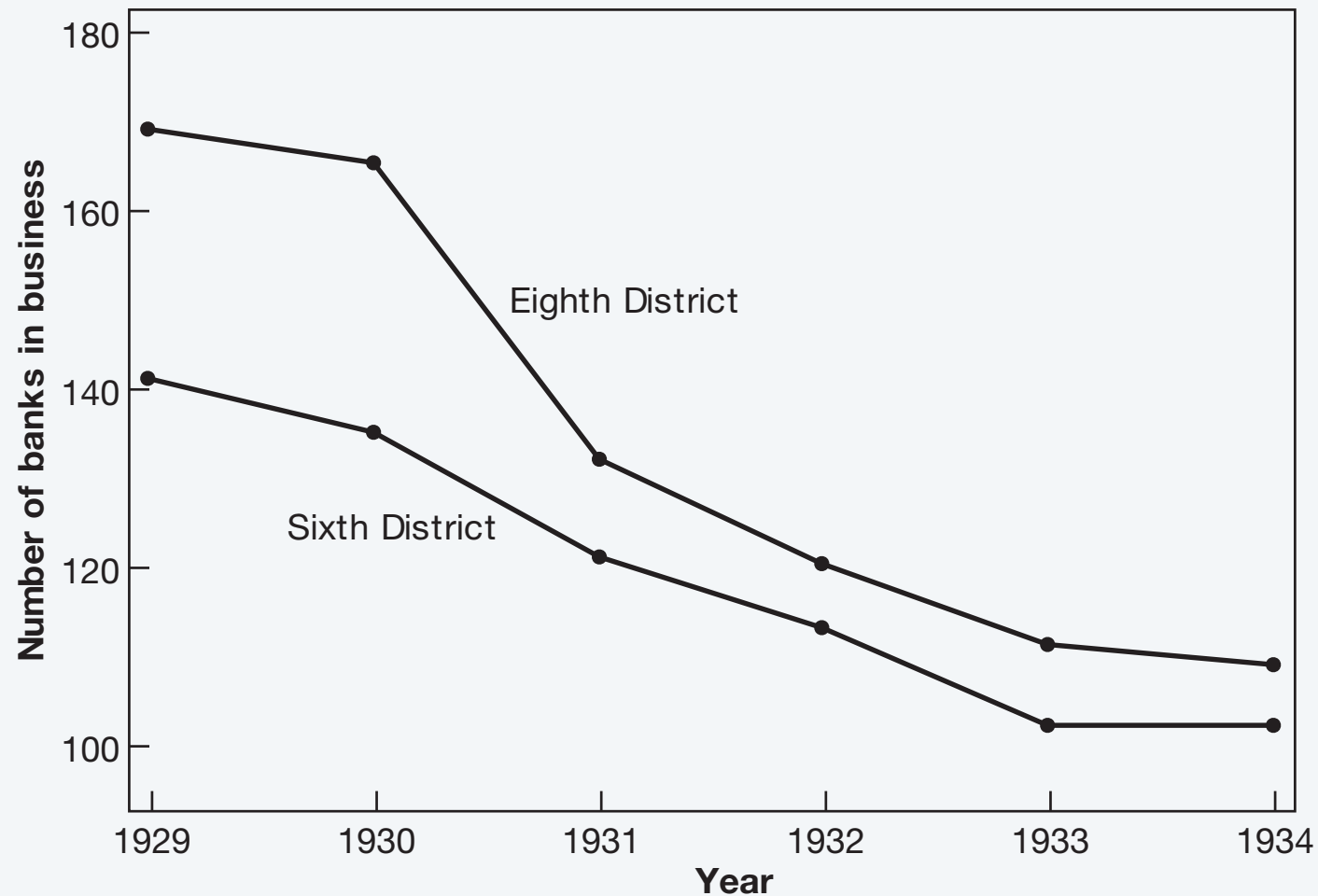
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- Why is the control group a good proxy for counterfactual? (when we do not require that treatment group and control group look the same before the treatment)
- Key assumption: **Common trends**
  - Absent any treatment, the trend in the treatment group is what we should have expected to see in the control group
- With more data, this assumption can be tested to some extent (although cannot be fully tested)



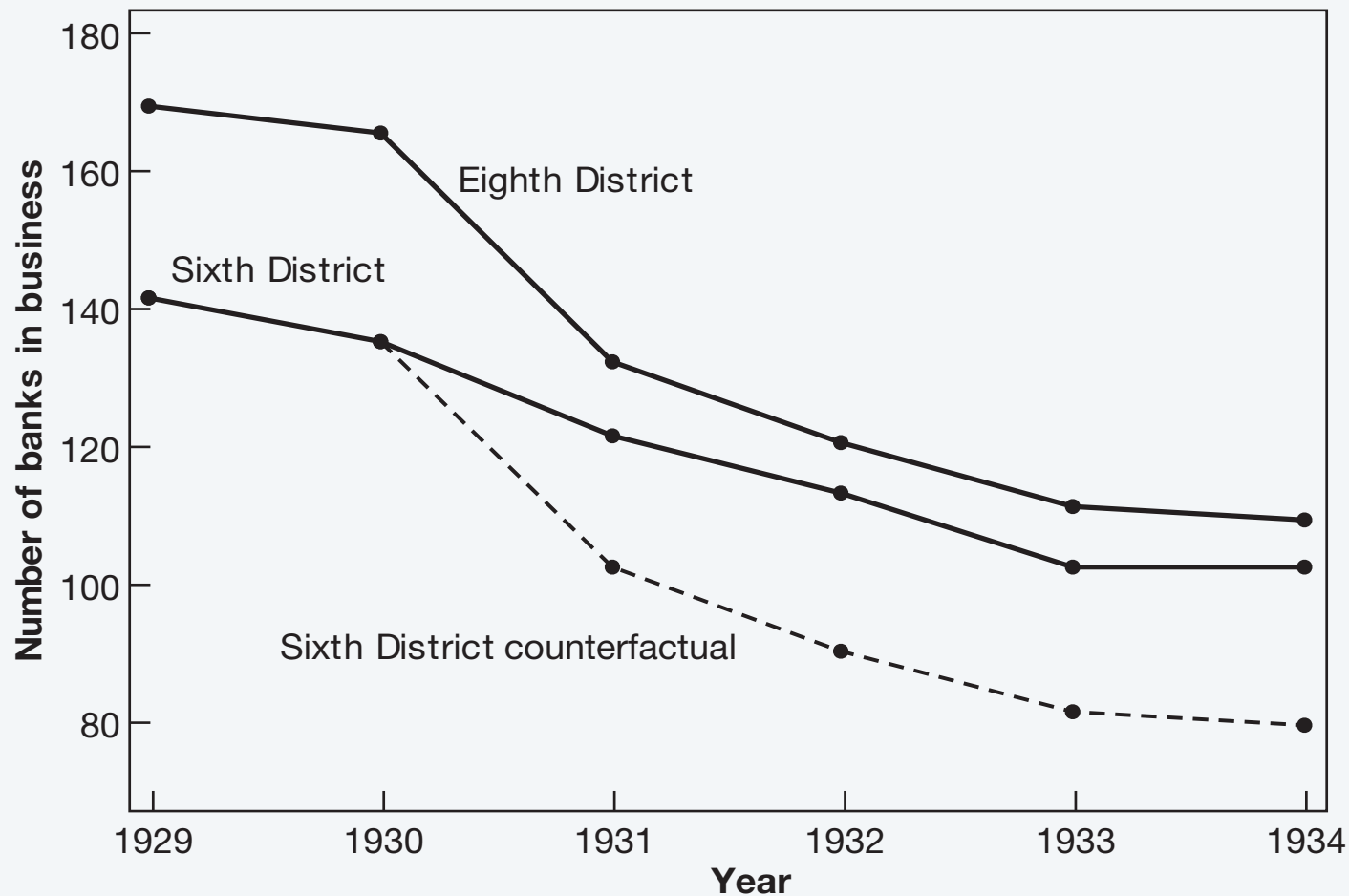
Parallel trends prior to the treatment (1929-1930)  
Parallel again with the same policy from 1932

FIGURE 5.2  
Trends in bank failures in the Sixth and Eighth Federal  
Reserve Districts



When pre-policy trends are parallel, treatment group's post-policy counterfactual is likely to have common trend as in the control group

FIGURE 5.3  
Trends in bank failures in the Sixth and Eighth Federal Reserve Districts, and the Sixth District's DD counterfactual



# Difference-in-Difference (DID) in regression

$$Y_{dt} = \alpha + \beta Treat_d + \gamma POST_t + \delta_{DD}(Treat_d * POST_t) + \epsilon_{dt}$$

- $\delta_{DD}$ : the causal effect of the treatment
- $\beta$ : pre-existing difference between treatment and control
- $\gamma$ : time trend - changes in outcome over time regardless of treatment

# Minimal legal drinking age (MLDA) revisited

- All US states set the MLDA to 21 in 1988
- Before that, some states had lowered it to below 21
  - Alabama lowered to 19 in 1975
  - Alaska remained at 21
- Difference-in-Difference design:
  - Two groups
    - Treatment group: Alabama
    - Control group: Alaska
  - Two time periods
    - Before policy: before 1975
    - After policy after 1975

# $\delta_{DD}$ : the causal effect of the MLDA

TABLE 5.2  
Regression DD estimates of MLDA effects on death rates

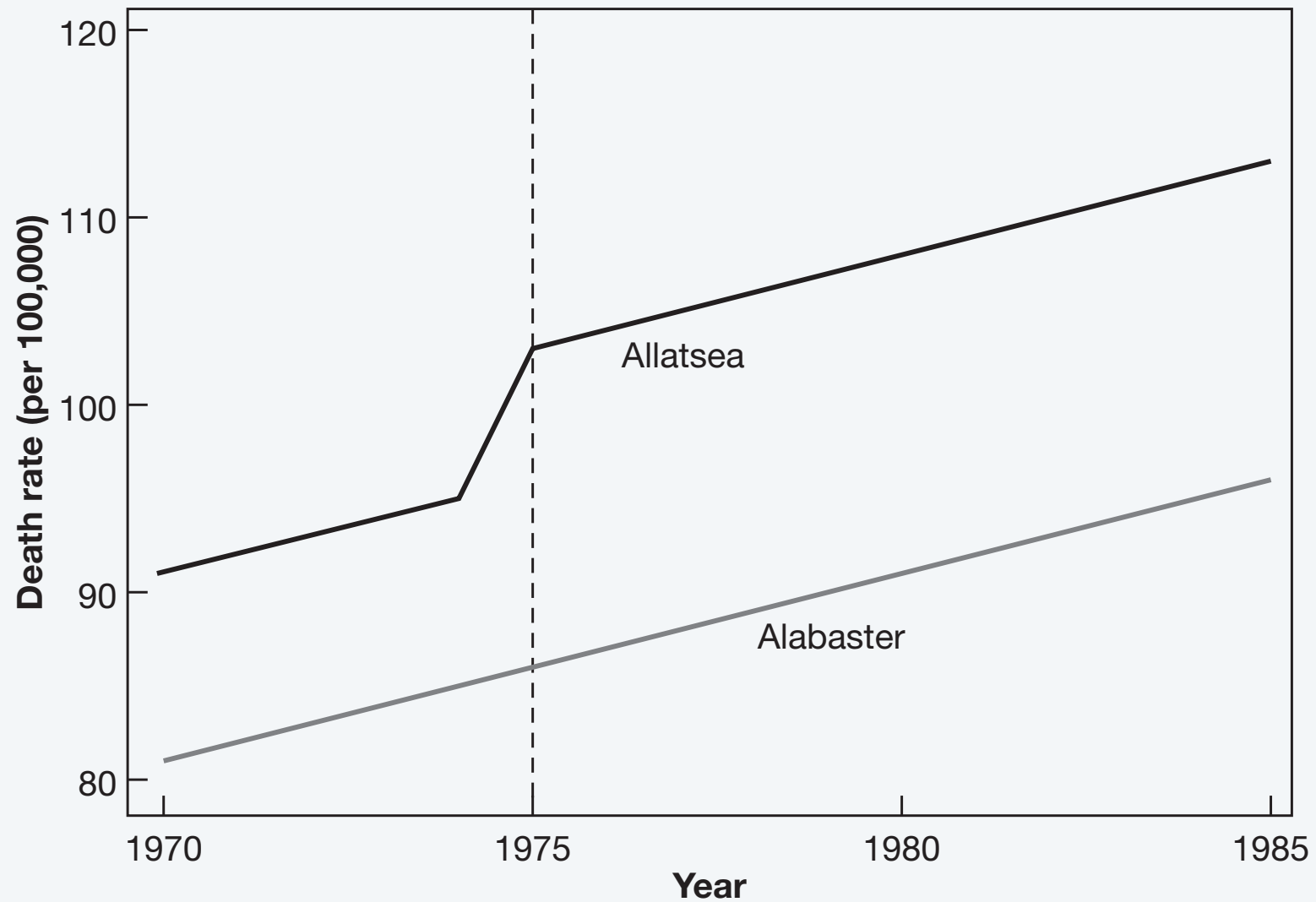
| Dependent variable      | (1)             | (2)            | (3)             | (4)            |
|-------------------------|-----------------|----------------|-----------------|----------------|
| All deaths              | 10.80<br>(4.59) | 8.47<br>(5.10) | 12.41<br>(4.60) | 9.65<br>(4.64) |
| Motor vehicle accidents | 7.59<br>(2.50)  | 6.64<br>(2.66) | 7.50<br>(2.27)  | 6.46<br>(2.24) |
| Suicide                 | .59<br>(.59)    | .47<br>(.79)   | 1.49<br>(.88)   | 1.26<br>(.89)  |
| All internal causes     | 1.33<br>(1.59)  | .08<br>(1.93)  | 1.89<br>(1.78)  | 1.28<br>(1.45) |
| State trends            | No              | Yes            | No              | Yes            |
| Weights                 | No              | No             | Yes             | Yes            |

# Is the DID assumption valid?

Are pre-policy trends parallel between the treatment and control groups?

- Consider different scenarios for two fake states
  - Treatment group: Allatsea (reduced MLDA to 18)
  - Control group: Alabaster (remained at 21)

FIGURE 5.4  
An MLDA effect in states with parallel trends



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FIGURE 5.5  
A spurious MLDA effect in states where trends are not parallel

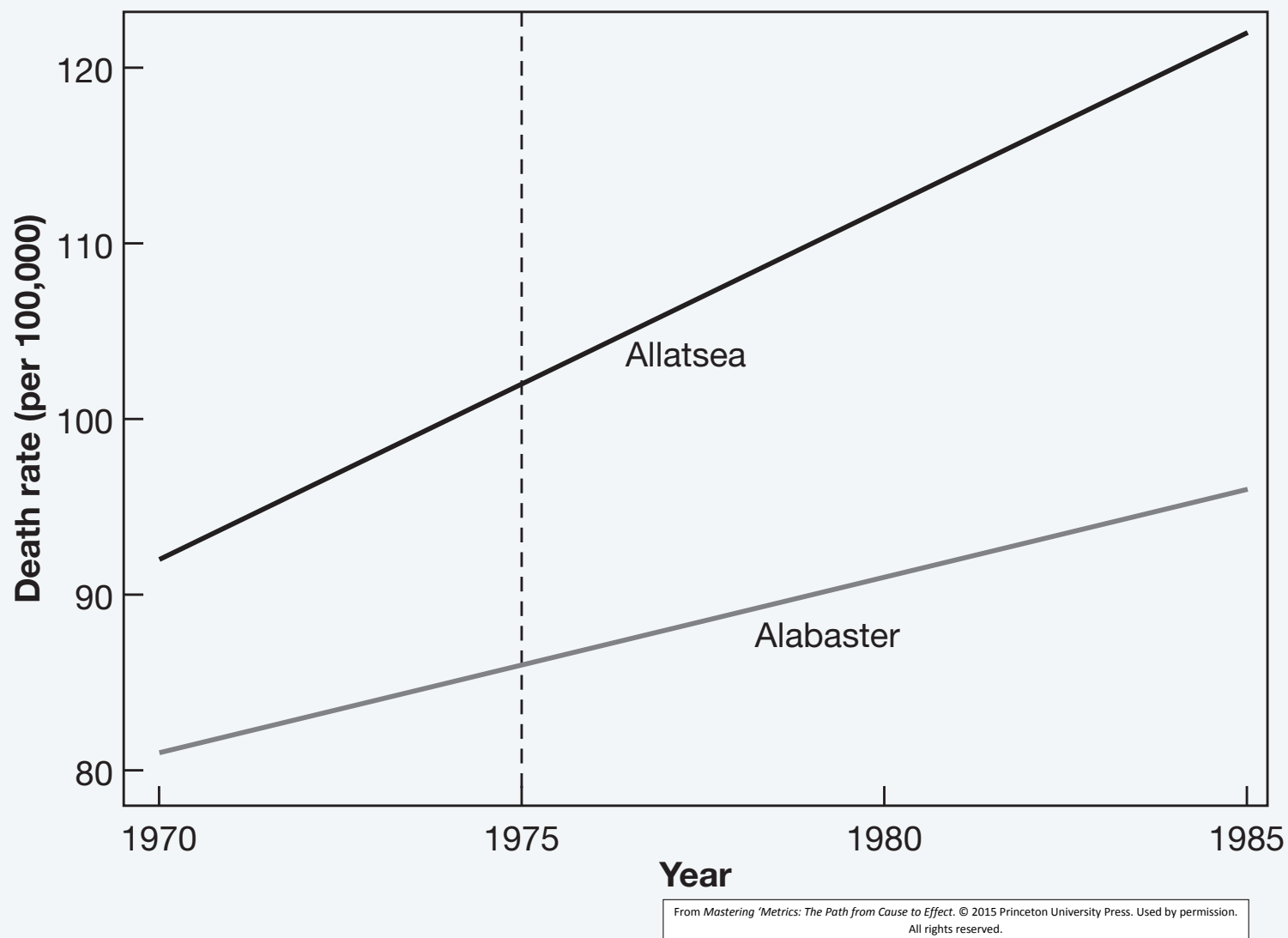
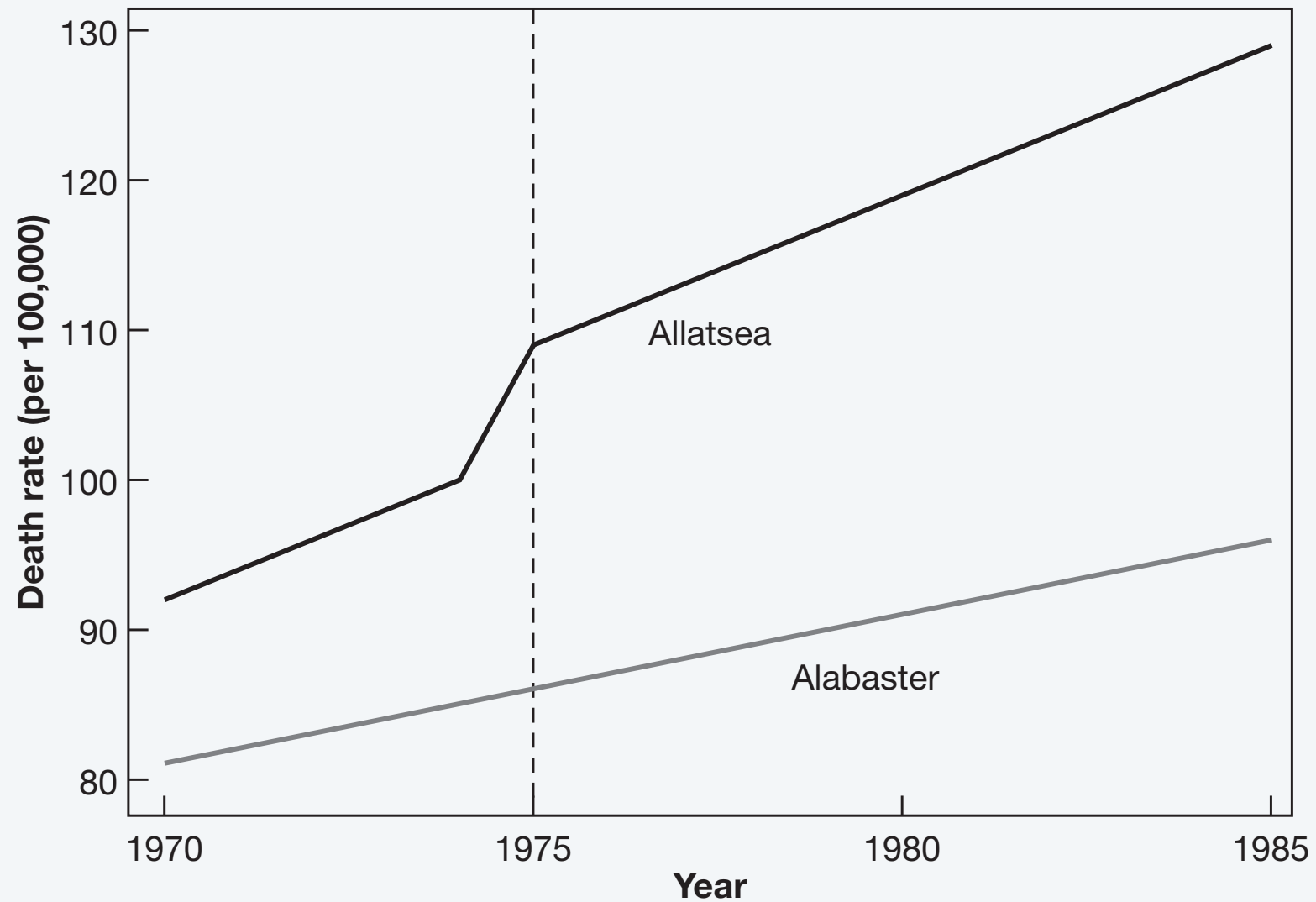




FIGURE 5.6  
A real MLDA effect, visible even though trends are not parallel



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# Defending the common trends assumption

Two approaches:

1. A compelling graph showing parallel trends before the treatment
2. If not compelling, control for group-specific time trends directly
  - For example, include an interaction term of Allatsea\*year in the DID regression

# What else could be a source of bias?

# What else could be a source of bias?

Other policies that happen at the same time and are also relevant to outcomes

- Price of a drink: beer taxes
  - States might increase beer taxes at the same time that they increase their MLDA (broader effort to reduce drinking)
- What to do?
  - Control for changes in beer tax rates

TABLE 5.3  
Regression DD estimates of MLDA effects controlling for beer taxes

| Dependent variable         | Without trends        |                 | With trends           |                   |
|----------------------------|-----------------------|-----------------|-----------------------|-------------------|
|                            | Fraction legal<br>(1) | Beer tax<br>(2) | Fraction legal<br>(3) | Beer tax<br>(4)   |
| All deaths                 | 10.98<br>(4.69)       | 1.51<br>(9.07)  | 10.03<br>(4.92)       | −5.52<br>(32.24)  |
| Motor vehicle<br>accidents | 7.59<br>(2.56)        | 3.82<br>(5.40)  | 6.89<br>(2.66)        | 26.88<br>(20.12)  |
| Suicide                    | .45<br>(.60)          | −3.05<br>(1.63) | .38<br>(.77)          | −12.13<br>(8.82)  |
| Internal causes            | 1.46<br>(1.61)        | −1.36<br>(3.07) | .88<br>(1.81)         | −10.31<br>(11.64) |

# Takeaways

- Treatment group and control do not have to look the same prior to the treatment
- Start with a simple 2-by-2 DD table
- Assess common trends visually
  - Trends of the treatment group and control group should look parallel before treatment
  - Treatment effects should be a level shift after the treatment
- Are there other concurrent policies that might also affect the outcomes?
  - Try to control for other concurrent policy changes in the regression

ANY  
QUESTIONS  
?